

Embedded AI – Assignment 4

EMBD X424

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https://github.com/electrickitty2/embedx424_assignment_4

Garbage Classification Tiny CNN — All TFLite Quantization Options

- Download or load a dataset
- Train a tiny grayscale CNN classifier
- Export all five TFLite quantization variants:
 - Float32
 - Dynamic-range Quantization
 - Float16
 - Full int8, float I/O
 - Full int8 strict int8 I/O
- Sanity-check every exported model on desktop with `tf.lite.interpreter`

Definitions

- Baseline – Just add the export for all of the different models before changing any parameters
- Final – This is the model with some tweaks to make it more accurate

Changes made to make Baseline

- Added Export of various model quantization
- Added Accuracy and Loss graphs

Baseline Model

... Model: "tiny_ds_cnn"

Layer (type)	Output Shape	Param #
input_layer (InputLayer)	(None, 64, 64, 1)	0
conv2d (Conv2D)	(None, 64, 64, 16)	160
max_pooling2d (MaxPooling2D)	(None, 32, 32, 16)	0
depthwise_conv2d (DepthwiseConv2D)	(None, 32, 32, 16)	160
conv2d_1 (Conv2D)	(None, 32, 32, 24)	408
max_pooling2d_1 (MaxPooling2D)	(None, 16, 16, 24)	0
depthwise_conv2d_1 (DepthwiseConv2D)	(None, 16, 16, 24)	240
conv2d_2 (Conv2D)	(None, 16, 16, 32)	800
max_pooling2d_2 (MaxPooling2D)	(None, 8, 8, 32)	0
global_average_pooling2d (GlobalAveragePooling2D)	(None, 32)	0
dense (Dense)	(None, 6)	198

Total params: 1,966 (7.68 KB)

Trainable params: 1,966 (7.68 KB)

Non-trainable params: 0 (0.00 B)

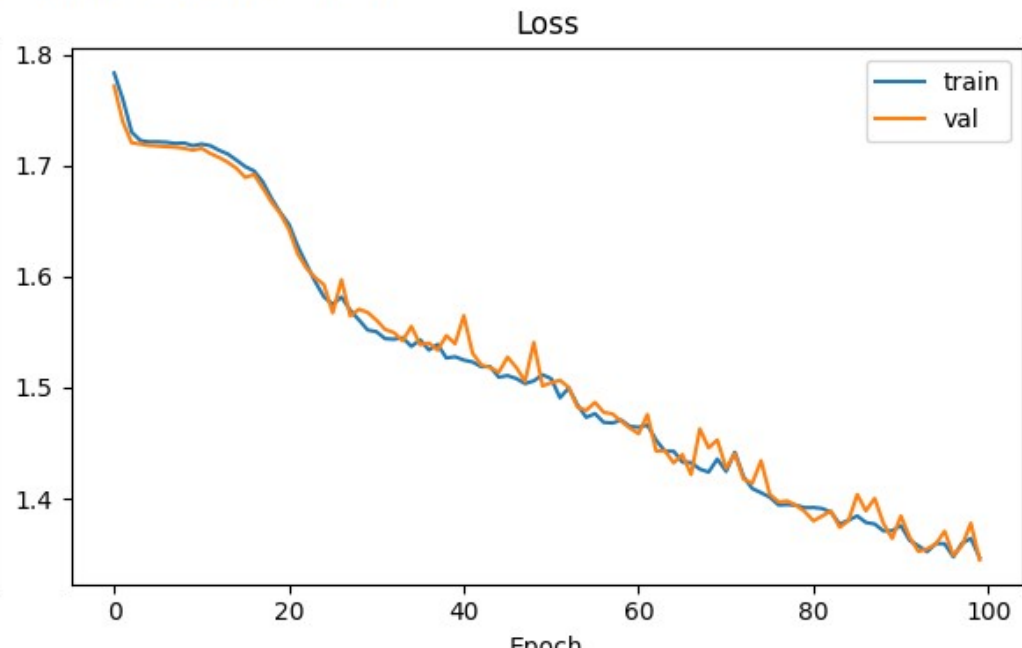
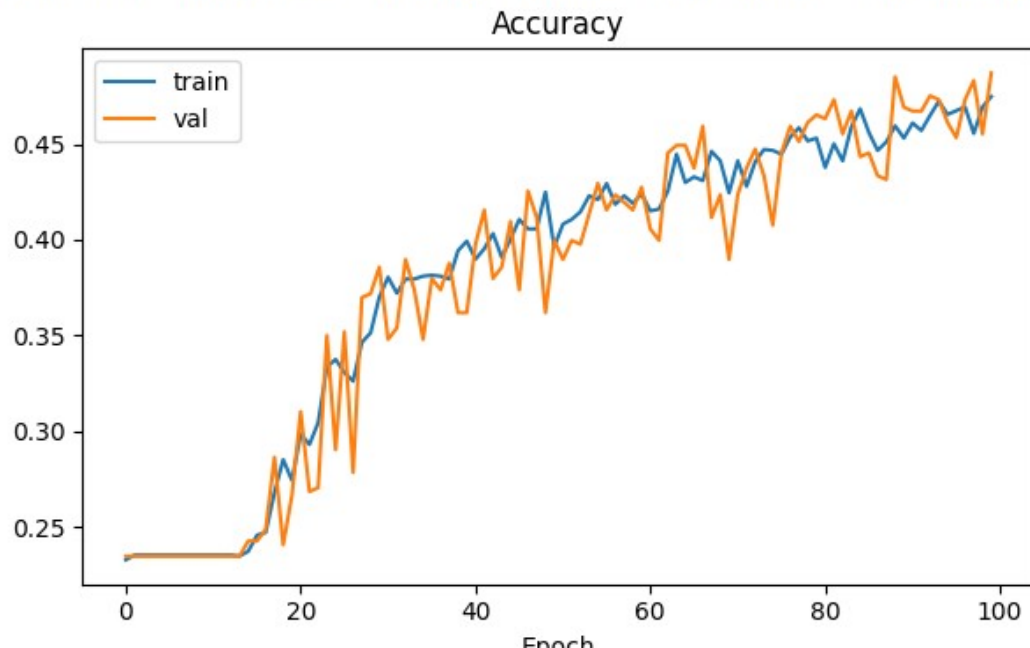
Baseline results

Ground truth: cardboard

Model	In dtype	Out dtype	Pred	Conf	Size	Runs
1 float32	float32	float32	cardboard	40.5% ✓	16K	No
2 dynamic-range	float32	float32	cardboard	40.5% ✓	16K	No
3 float16	float32	float32	cardboard	40.5% ✓	12K	No
4 int8 float I/O	float32	float32	cardboard	41.0% ✓	12K	No
5 int8 strict I/O	int8	int8	cardboard	41.0% ✓	12K	No

Baseline training vs validation

```
Epoch 97/100  
32/32 - 3s - 89ms/step - accuracy: 0.4674 - loss: 1.3592 - val_accuracy: 0.4533 - val_loss: 1.3705  
Epoch 98/100  
32/32 - 1s - 47ms/step - accuracy: 0.4694 - loss: 1.3474 - val_accuracy: 0.4732 - val_loss: 1.3488  
Epoch 99/100  
32/32 - 2s - 63ms/step - accuracy: 0.4555 - loss: 1.3597 - val_accuracy: 0.4831 - val_loss: 1.3576  
Epoch 100/100  
32/32 - 2s - 67ms/step - accuracy: 0.4694 - loss: 1.3641 - val_accuracy: 0.4553 - val_loss: 1.3778  
Epoch 100/100  
32/32 - 2s - 48ms/step - accuracy: 0.4748 - loss: 1.3468 - val_accuracy: 0.4871 - val_loss: 1.3447
```



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Final Model

... Model: "tiny_ds_cnn"

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conv2d_1 (Conv2D)	(None, 32, 32, 24)	408
max_pooling2d (MaxPooling2D)	(None, 16, 16, 24)	0
depthwise_conv2d_1 (DepthwiseConv2D)	(None, 16, 16, 24)	240
conv2d_2 (Conv2D)	(None, 16, 16, 32)	800
max_pooling2d_1 (MaxPooling2D)	(None, 8, 8, 32)	0
global_average_pooling2d (GlobalAveragePooling2D)	(None, 32)	0
dense (Dense)	(None, 6)	198

Total params: 1,966 (7.68 KB)

Trainable params: 1,966 (7.68 KB)

Non-trainable params: 0 (0.00 B)

Final Results

Ground truth: cardboard

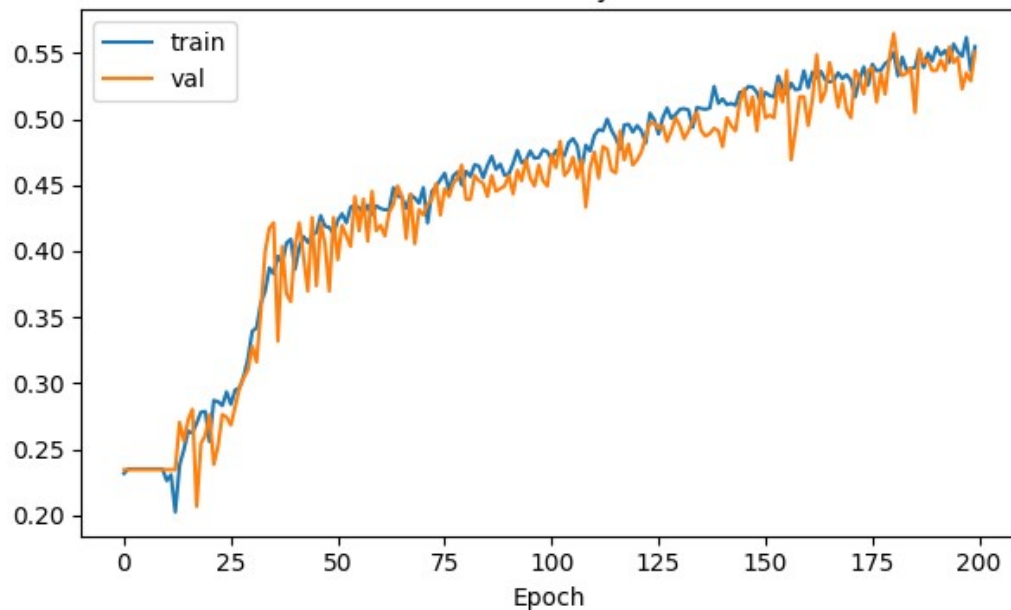
Model	In dtype	Out dtype	Pred	Conf	Size	Runs

1 float32	float32	float32	cardboard	62.1% ✓	16K	Yes 23fps
2 dynamic-range	float32	float32	cardboard	62.1% ✓	16K	Yes 23fps
3 float16	float32	float32	cardboard	62.2% ✓	12K	No
4 int8 float I/O	float32	float32	cardboard	64.5% ✓	12K	No
5 int8 strict I/O	int8	int8	cardboard	62.9% ✓	12K	No

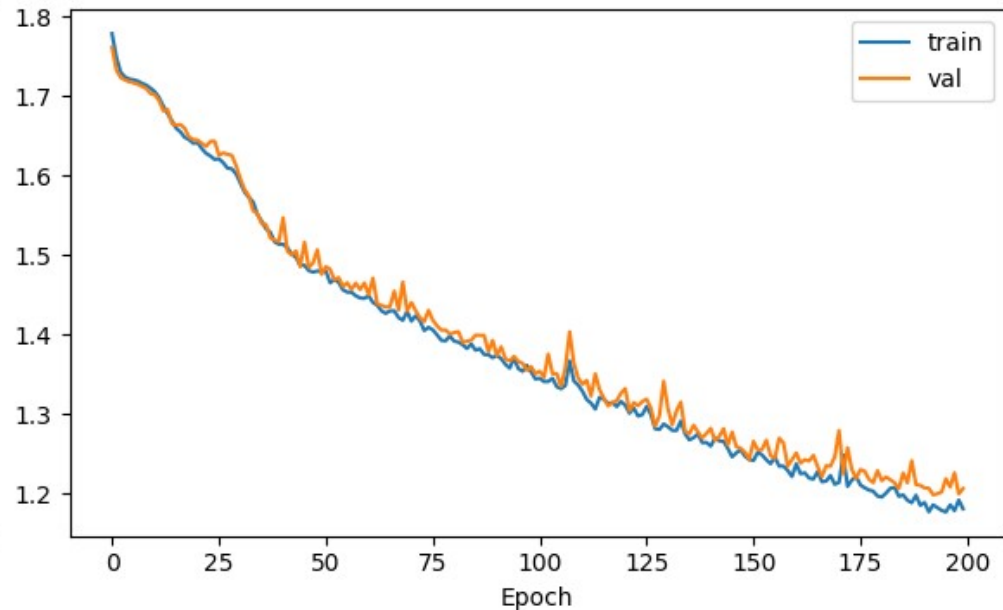
Final Training vs Validation

```
32/32 - 3s - 100ms/step - accuracy: 0.5509 - loss: 1.1774 - val_accuracy: 0.5467 - val_loss: 1.2192
Epoch 197/200
32/32 - 2s - 71ms/step - accuracy: 0.5474 - loss: 1.1864 - val_accuracy: 0.5229 - val_loss: 1.2087
Epoch 198/200
32/32 - 2s - 72ms/step - accuracy: 0.5618 - loss: 1.1791 - val_accuracy: 0.5348 - val_loss: 1.2267
Epoch 199/200
32/32 - 2s - 72ms/step - accuracy: 0.5366 - loss: 1.1924 - val_accuracy: 0.5288 - val_loss: 1.2001
Epoch 200/200
32/32 - 2s - 65ms/step - accuracy: 0.5548 - loss: 1.1815 - val_accuracy: 0.5507 - val_loss: 1.2070
```

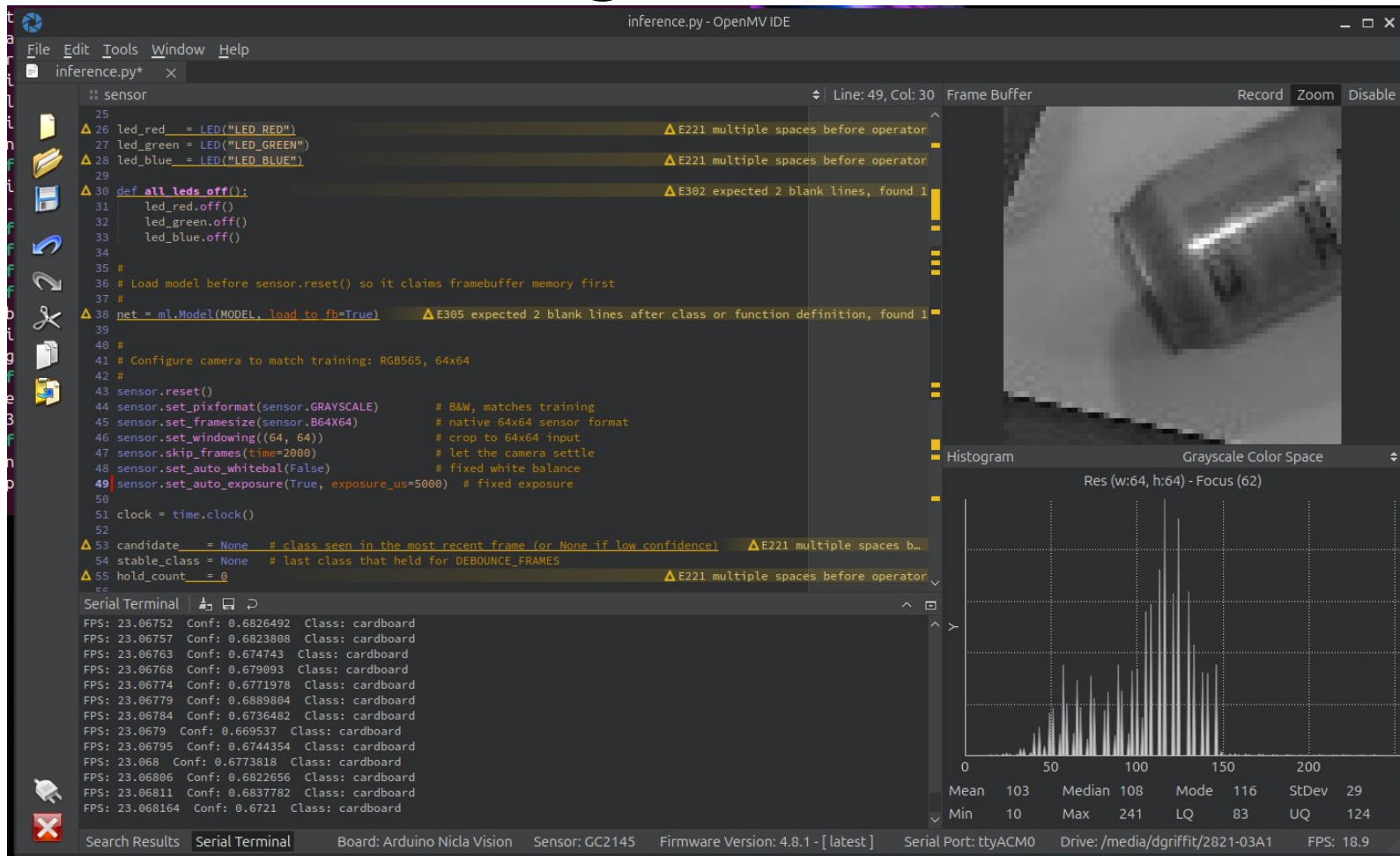
Accuracy



Loss



The can is recognized as cardboard



The screenshot displays the OpenMV IDE interface with the following components:

- Code Editor:** Contains a Python script for object detection. The script uses a sensor to capture a grayscale image, processes it with a neural network model, and outputs the detected class and confidence. The script includes comments and error messages.
- Frame Buffer:** Shows a grayscale image of a can, which is the input to the object detection model.
- Histogram:** A graph showing the distribution of pixel intensities in the grayscale image. The x-axis represents pixel intensity (0 to 255) and the y-axis represents frequency. The histogram is titled "Histogram" and "Grayscale Color Space".
- Serial Terminal:** Displays the output of the script, showing the detected class and confidence for each frame. The output is as follows:

FPS	Conf	Class
23.06752	0.6826492	cardboard
23.06757	0.6823808	cardboard
23.06763	0.674743	cardboard
23.06768	0.679093	cardboard
23.06774	0.6771978	cardboard
23.06779	0.6889804	cardboard
23.06784	0.6736482	cardboard
23.0679	0.669537	cardboard
23.06795	0.6744354	cardboard
23.068	0.6773818	cardboard
23.06806	0.6822656	cardboard
23.06811	0.6837782	cardboard
23.068164	0.6721	cardboard

The bottom status bar shows the following information:

- Search Results
- Serial Terminal
- Board: Arduino Nicla Vision
- Sensor: GC2145
- Firmware Version: 4.8.1 - [latest]
- Serial Port: ttyACM0
- Drive: /media/dgriffit/2821-03A1
- FPS: 18.9

Summary – What I learned

- The biggest learning for me was that the smaller model does not necessarily mean it will fit.
- Removing the pooling seems to have stuck the inference on cardboard, trash, sometimes metal.